Project Design Phase-II
Technology Stack (Architecture & Stack)

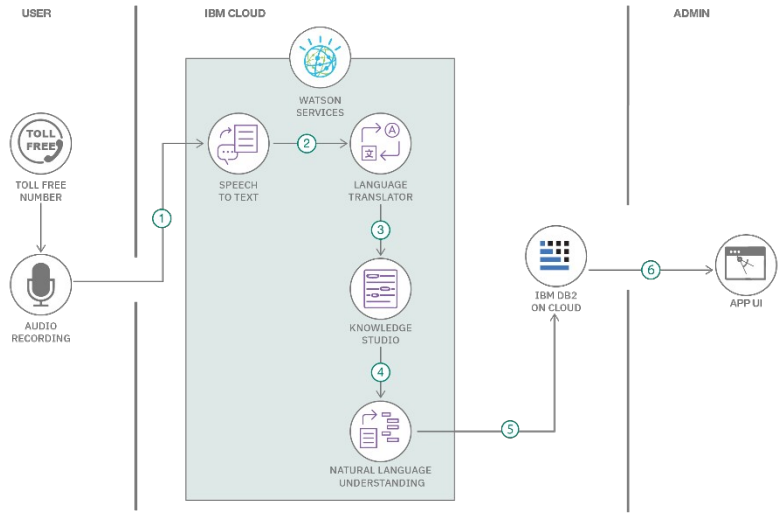
Date	22.09.2026
Team ID	SWTID-2026-7438
Project Name	AI FAQ Assistant
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: AI FAQ Assistant architecture

Reference: Architecture designed for a web-based AI FAQ Assistant with application, FAQ database and AI/NLP service.



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web interface for users to ask questions, search FAQs and view answers.	HTML, CSS, JavaScript
2	Application Logic-1	Handles user registration, login, FAQ search and session management.	Java / Spring Boot
3	Application Logic-2	Processes natural-language questions and retrieves relevant FAQ answers.	Python / NLP
4	Application Logic-3	AI conversational assistance and response generation.	AI/NLP model or chatbot service
5	Database	Stores user accounts, FAQ records, categories and feedback.	MySQL / PostgreSQL
6	Cloud Database	Cloud-hosted database service for scalable application data.	Cloud SQL / Managed Database
7	File Storage	Stores supporting documents or FAQ reference files.	Cloud Object Storage / Local Filesystem
8	External API-1	Authentication or email service for registration and notifications.	Gmail / Email API
9	External API-2	Optional AI/NLP service for question understanding.	AI/NLP API
10	Machine Learning Model	Understands user questions and identifies relevant FAQ content.	NLP / Embedding-based Retrieval Model
11	Infrastructure (Server / Cloud)	Application deployment and hosting environment.	Cloud Server / Container Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Use open-source frameworks for the web application, backend and NLP components.	Spring Boot, Python, Flask/FastAPI
2	Security Implementations	Authentication, authorization, password protection and secure API communication.	HTTPS, Encryption, JWT, OWASP practices
3	Scalable Architecture	Separate UI, application logic, AI/NLP and database layers for scalability.	3-tier architecture / Microservices
4	Availability	Use reliable cloud hosting, database backups and service monitoring.	Cloud hosting, Load Balancer, Monitoring
5	Performance	Optimize FAQ retrieval, use indexing/cache and reduce unnecessary model calls.	Database Indexing, Cache, API Optimization

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d